STUDENT PERFORMANCE ANALYTICS SYSTEM

Python Mini Project Documentation

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Organization: EWB Courses
Project Title: Student Performance Analytics System
Technologies Used: Python, NumPy, Pandas
Academic Year: 2026

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1. ABSTRACT

The Student Performance Analytics System is a Python-based mini project developed to analyze student academic performance using marks and attendance data. The system uses Pandas to store and process student records in a DataFrame and NumPy to perform numerical calculations.

The project calculates total marks and average marks for each student, assigns grades, identifies pass and fail results, and determines the highest- and lowest-performing students. It also performs subject-wise and department-wise performance analysis, identifies the top five students, and highlights students with attendance below 75%.

This project demonstrates the practical application of Python programming, functions, conditional statements, loops, NumPy, and Pandas in data analysis.

2. INTRODUCTION

Student performance analysis helps educational institutions understand academic progress and identify areas that require improvement. Manually calculating marks, grades, and averages for many students can be time-consuming.

The Student Performance Analytics System simplifies this process by automatically analyzing student records using Python. It processes data from a CSV file and displays meaningful results in the terminal.

The project focuses on basic data analytics concepts and provides a simple way to summarize student performance.

3. OBJECTIVES

The main objectives of this project are:

1. To maintain and analyze student records using a CSV dataset.
2. To calculate total marks and average marks for each student.
3. To assign grades based on average marks.
4. To classify students as pass or fail.
5. To identify the highest- and lowest-performing students.
6. To calculate subject-wise average marks.
7. To identify the top five performing students.
8. To analyze student attendance and identify low-attendance students.
9. To compare academic performance across departments.
10. To apply Python, NumPy, and Pandas to a practical data analysis problem.

4. TECHNOLOGIES USED

4.1 Python

Python is the main programming language used to develop the project. It is used for calculations, conditional statements, loops, functions, and displaying results.

4.2 Pandas

Pandas is used to read the CSV dataset, store student records in a DataFrame, calculate new columns, group students by department, sort records, and filter students based on attendance.

4.3 NumPy

NumPy is used for numerical calculations, including calculating average marks and overall attendance.

4.4 CSV

CSV stands for Comma-Separated Values. The `students.csv` file stores student details, department names, subject marks, and attendance percentages.

4.5 Visual Studio Code

Visual Studio Code is used to write, save, and execute the Python program.

5. PROJECT REQUIREMENTS

Hardware Requirements

- Computer or laptop

- Minimum 4 GB RAM recommended
- Sufficient storage for Python, libraries, and project files

Software Requirements

- Windows or another operating system that supports Python
- Python 3.8.10
- Visual Studio Code
- Pandas 2.0.3
- NumPy 1.24.4

6. DATASET DESCRIPTION

The project uses a CSV file named `students.csv`, containing 20 student records.

The dataset includes the following columns:

Column	Description
Student_ID	Unique identification number for each student
Name	Name of the student
Department	Department to which the student belongs
Python	Marks obtained in Python
SQL	Marks obtained in SQL
Maths	Marks obtained in Mathematics
Attendance	Student attendance percentage

The dataset contains students from different departments, including CAI, CSE, CSD, ECE, EEE, MECH, and CSM.

The data is loaded into a Pandas DataFrame using the `pd.read_csv()` function.

7. METHODOLOGY

The project follows these steps:

1. Load the student dataset from `students.csv`.
2. Display the student records.
3. Calculate total marks and average marks.
4. Assign grades using a user-defined function.
5. Determine pass or fail results.
6. Calculate overall class performance.
7. Analyze subject-wise averages.
8. Sort student records to identify the top five performers.
9. Analyze attendance percentages.
10. Calculate department-wise average marks.

11. Display the results in the terminal.

8. IMPLEMENTATION

8.1 Loading the Dataset

The program imports Pandas and NumPy and loads student records from the CSV file.

```
import pandas as pd
import numpy as np

students = pd.read_csv("students.csv")
```

Pandas reads the CSV file and stores the data in a DataFrame for further analysis.

8.2 Calculating Total and Average Marks

The program calculates the total marks by adding the marks obtained in Python, SQL, and Maths.

Formula:

Total Marks = Python + SQL + Maths

Average Marks = Total Marks / 3

These calculations help evaluate the academic performance of each student.

8.3 Grade Calculation

A user-defined function named `calculate_grade()` assigns grades based on average marks.

The grading criteria are:

Average Marks	Grade
90 and above	A
80 to below 90	B
70 to below 80	C
60 to below 70	D
Below 60	F

The function uses conditional statements to determine the appropriate grade.

8.4 Pass and Fail Analysis

The `calculate_result()` function classifies students according to their average marks.

- Average marks of 40 or above: Pass
- Average marks below 40: Fail

The program counts the number of passed and failed students using Pandas operations.

8.5 Class Performance Analysis

The program calculates the overall class average using NumPy. It also identifies the highest- and lowest-performing students using the maximum and minimum average marks.

This analysis provides a general overview of student performance.

8.6 Subject-Wise Performance Analysis

The program calculates the average marks for Python, SQL, and Maths.

A loop processes each subject and stores its average in a dictionary. The program then identifies the subject with the highest average marks.

8.7 Top Five Performing Students

The program sorts student records in descending order based on average marks and displays the top five students.

This helps identify students with the strongest academic performance.

8.8 Attendance Analysis

The program calculates the overall average attendance percentage and identifies students whose attendance is below 75%.

This analysis helps highlight students who may require attention regarding attendance.

8.9 Department-Wise Performance Analysis

The program groups students by department and calculates the average marks for each department using Pandas `groupby()`.

The department with the highest average marks is identified as the best-performing department in the dataset.

9. FEATURES OF THE SYSTEM

The main features of the Student Performance Analytics System are:

- Student dataset loading and display
- Automatic total and average marks calculation
- Grade assignment
- Pass and fail classification
- Overall class performance summary
- Highest- and lowest-performing student identification
- Subject-wise average analysis
- Top five student ranking
- Attendance percentage analysis
- Identification of students with attendance below 75%
- Department-wise performance comparison

10. PYTHON CONCEPTS USED

The project demonstrates the following programming concepts:

Variables and data types: Used to store student information and calculated values.

Conditional statements: Used for assigning grades and determining pass or fail results.

Functions: Used to organize reusable logic for grade and result calculation.

Loops: Used to process subject-wise and department-wise analysis.

Lists and dictionaries: Used to organize subject names and store subject averages.

NumPy: Used for numerical calculations and averages.

Pandas DataFrame: Used for loading, filtering, sorting, and grouping student data.

11. RESULTS AND OUTPUT ANALYSIS

The program processes all 20 student records and produces the following results based on the current dataset and implemented rules.

11.1 Overall Class Performance

- Total number of students: 20
- Overall class average: 72.20
- Highest-performing student: Preethi
- Highest average marks: 93.67
- Lowest-performing student: Mounish
- Lowest average marks: 38.33

11.2 Pass and Fail Results

- Total passed students: 18
- Total failed students: 2

The pass and fail results are based on the project rule that an average of at least 40 marks is considered a pass.

11.3 Subject-Wise Results

Subject	Average Marks
Python	70.75
SQL	72.75
Maths	73.10

Maths has the highest average marks among the three subjects.

11.4 Attendance Results

- Overall average attendance: 82.90%
- Students with attendance below 75%: 4

11.5 Department-Wise Results

Department	Average Marks
CAI	81.44
CSD	87.78
CSE	60.00
CSM	74.67
ECE	59.56
EEE	88.56
MECH	44.00

Based on the current dataset, EEE is the best-performing department, with an average of 88.56 marks.

11.6 Top Five Performing Students

The program ranks students according to their average marks. The highest-performing students are:

1. Preethi
2. Sukitha
3. Akshaya
4. Manohar
5. Harika

These results reflect the current contents of `students.csv`. If the dataset or calculation rules are changed, the output may also change.

12. OUTPUT SCREENSHOTS

Figure 1: Student Records

```
===== STUDENT PERFORMANCE ANALYTICS SYSTEM =====

Student Records:
 Student_ID     Name Department  Python  SQL  Maths  Attendance
        101   Balaji        CAI      85   78     90          92
        102    Sujan        CAI      65   72     68          80
        103  Akshaya        CAI      92   95     88          96
        104  Meghana        CSE      45   55     50          68
        105   Harika        CSD      88   91     84          94
        106   Ambika        ECE      72   68     75          82
        107  Preethi        EEE      95   89     97          98
        108  Mounish       MECH      38   42     35          60
        109   Kaveri        CSM      78   82     80          88
        110  Bhuvana        CSE      56   60     52          75
        111   Tharun        CSD      84   86     90          91
        112     Prem        ECE      67   73     64          79
        113  Sukitha        EEE      91   94     93          97
        114  Chaitra       MECH      48   46     55          70
        115 Harshini        CSM      80   85     87          90
        116   Rajeev        CSE      74   70     78          84
        117  Manohar        CSD      89   92     86          95
        118  Muneesh        ECE      32   40     45          55
        119  Likitha        EEE      76   79     83          86
        120  Nikitha        CSM      60   58     62          78

Total Number of Students: 20

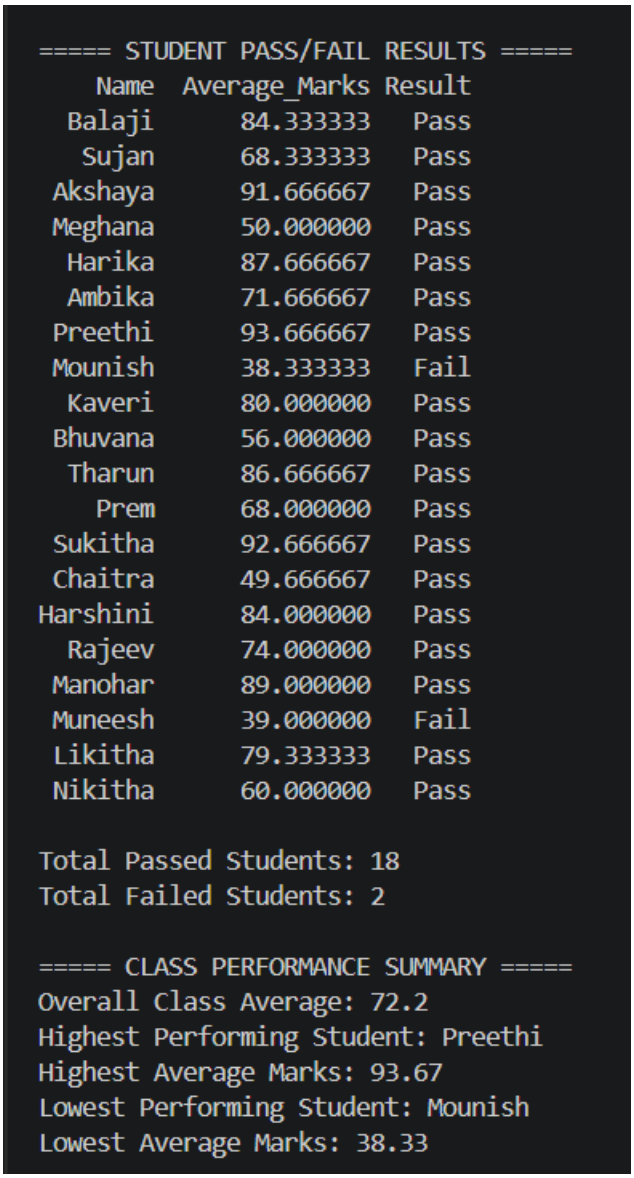
Department Names:
['CAI' 'CSE' 'CSD' 'ECE' 'EEE' 'MECH' 'CSM']
```

Figure 2: Student Marks and Grades

```
===== STUDENT MARKS ANALYSIS =====
    Name Department  Total_Marks  Average_Marks
  Balaji        CAI          253      84.333333
   Sujan        CAI          205      68.333333
 Akshaya        CAI          275      91.666667
 Meghana        CSE          150      50.000000
  Harika        CSD          263      87.666667
  Ambika        ECE          215      71.666667
 Preethi        EEE          281      93.666667
 Mounish       MECH          115      38.333333
  Kaveri        CSM          240      80.000000
 Bhuvana        CSE          168      56.000000
  Tharun        CSD          260      86.666667
    Prem        ECE          204      68.000000
 Sukitha        EEE          278      92.666667
 Chaitra       MECH          149      49.666667
Harshini        CSM          252      84.000000
  Rajeev        CSE          222      74.000000
 Manohar        CSD          267      89.000000
 Muneesh        ECE          117      39.000000
 Likitha        EEE          238      79.333333
 Nikitha        CSM          180      60.000000
```

```
===== STUDENT GRADES =====
    Name  Average_Marks Grade
  Balaji      84.333333     B
   Sujan      68.333333     D
 Akshaya      91.666667     A
 Meghana      50.000000     F
  Harika      87.666667     B
  Ambika      71.666667     C
 Preethi      93.666667     A
 Mounish      38.333333     F
  Kaveri      80.000000     B
 Bhuvana      56.000000     F
  Tharun      86.666667     B
    Prem      68.000000     D
 Sukitha      92.666667     A
 Chaitra      49.666667     F
Harshini      84.000000     B
  Rajeev      74.000000     C
 Manohar      89.000000     B
 Muneesh      39.000000     F
 Likitha      79.333333     C
 Nikitha      60.000000     D
```

Figure 3: Pass/Fail Results and Class Summary



```
===== STUDENT PASS/FAIL RESULTS =====
    Name  Average_Marks Result
  Balaji      84.333333   Pass
   Sujan      68.333333   Pass
 Akshaya      91.666667   Pass
 Meghana      50.000000   Pass
  Harika      87.666667   Pass
  Ambika      71.666667   Pass
 Preethi      93.666667   Pass
 Mounish      38.333333   Fail
  Kaveri      80.000000   Pass
 Bhuvana      56.000000   Pass
  Tharun      86.666667   Pass
    Prem      68.000000   Pass
 Sukitha      92.666667   Pass
 Chaitra      49.666667   Pass
Harshini      84.000000   Pass
  Rajeev      74.000000   Pass
 Manohar      89.000000   Pass
 Muneesh      39.000000   Fail
 Likitha      79.333333   Pass
 Nikitha      60.000000   Pass

Total Passed Students: 18
Total Failed Students: 2

===== CLASS PERFORMANCE SUMMARY =====
Overall Class Average: 72.2
Highest Performing Student: Preethi
Highest Average Marks: 93.67
Lowest Performing Student: Mounish
Lowest Average Marks: 38.33
```

Figure 4: Subject-Wise Performance and Top Five Students

```
===== SUBJECT-WISE PERFORMANCE ANALYSIS =====
Python Average Marks: 70.75
SQL Average Marks: 72.75
Maths Average Marks: 73.1

Best Performing Subject: Maths
Highest Subject Average: 73.1

===== TOP 5 PERFORMING STUDENTS =====
   Name Department  Average_Marks Grade
Preethi        EEE      93.666667     A
Sukitha        EEE      92.666667     A
Akshaya        CAI      91.666667     A
Manohar        CSD      89.000000     B
 Harika        CSD      87.666667     B
```

Figure 5: Attendance Analysis

```
===== ATTENDANCE ANALYSIS =====
Overall Average Attendance: 82.9 %

Students with Attendance Below 75%:
   Name Department  Attendance
Meghana        CSE          68
Mounish       MECH          60
Chaitra       MECH          70
Muneesh        ECE          55

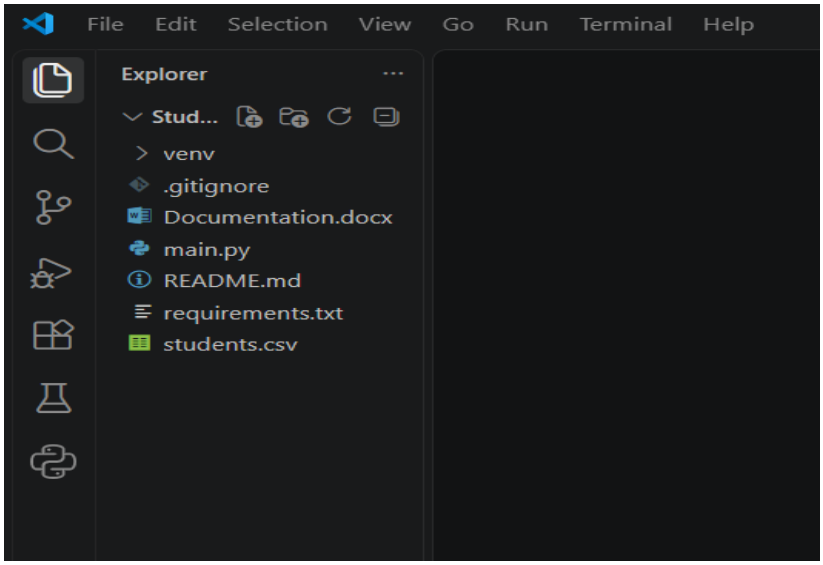
Total Students with Low Attendance: 4
```

Figure 6: Department-Wise Performance

```
===== DEPARTMENT-WISE PERFORMANCE =====
CAI Average Marks: 81.44
CSD Average Marks: 87.78
CSE Average Marks: 60.0
CSM Average Marks: 74.67
ECE Average Marks: 59.56
EEE Average Marks: 88.56
MECH Average Marks: 44.0

Best Performing Department: EEE
Department Average Marks: 88.56
```

Figure 7: Project Files in Visual Studio Code



13. TESTING

The program should be tested to confirm that the analysis works correctly.

Test Case	Expected Result
Load the CSV file	Student records are loaded successfully
Calculate total marks	Marks from all three subjects are added
Calculate average marks	Total marks are divided by three
Assign grades	Each student receives a grade based on the defined criteria
Calculate pass/fail	Students are classified according to the average marks rule
Find top performers	Students are sorted in descending order of average marks
Analyze attendance	Students below 75% attendance are identified
Analyze departments	Average marks are calculated for each department

The expected results should be compared with the actual terminal output during testing.

14. ADVANTAGES

- Reduces manual calculations.
- Processes multiple student records efficiently.
- Provides a clear summary of academic performance.
- Helps identify students who may need academic support.
- Demonstrates practical applications of Python and data analysis libraries.
- Can be extended to analyze larger datasets.

15. LIMITATIONS

- The current project uses a CSV file rather than a live database.
- The results depend on the accuracy of the input data.
- The grading and pass/fail criteria are predefined.
- The program displays results in the terminal and does not include a graphical user interface.
- The analysis uses three subjects and a fixed set of student records.

16. FUTURE ENHANCEMENTS

The project could be extended in the future by adding more subjects, supporting larger datasets, generating charts, exporting reports, or developing a graphical interface. These are possible improvements and are not part of the current project implementation.

17. CONCLUSION

The Student Performance Analytics System successfully demonstrates how Python, NumPy, and Pandas can be used to analyze student academic data. The project calculates total and average marks, assigns grades, determines pass and fail results, identifies top performers, analyzes attendance, and compares department-wise performance.

Through this project, practical knowledge of Python programming, functions, loops, conditional statements, numerical calculations, and data manipulation is applied to a real-world educational use case.

The project provides a foundation for understanding basic data analytics and can be enhanced with additional analytical features in the future.

18. REFERENCES

1. Python official documentation: https://docs.python.org/3/
2. Pandas official documentation: https://pandas.pydata.org/docs/
3. NumPy official documentation: https://numpy.org/doc/

19. PROJECT REPOSITORY

GitHub Repository URL: Student-Performance-Analytics-System

Or

https://github.com/akshaya615/Student-Performance-Analytics-System

Submission: The project source code, CSV dataset, README file, and documentation will be organized in the GitHub repository and submitted using the official EWB course submission form, as instructed by the trainer.